
THE ACUTE EFFECTS OF DYNAMIC AND BALLISTIC STRETCHING ON VERTICAL JUMP HEIGHT, FORCE, AND POWER

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ABSTRACT

Jaggers, JR, Swank, AM, Frost, KL, and Lee, CD. The acute effects of dynamic and ballistic stretching on vertical jump height, force, and power. *J Strength Cond Res* 22(6): 1844–1849, 2008—Stretching before performance is a common practice among athletes in hopes of increasing performance and reducing the risk of injury. However, cumulative results indicate a negative impact of static stretching and proprioceptive neuromuscular facilitation (PNF) on performance; thus, there is a need for evaluating other stretching strategies for effective warm-up. The purpose of this study was to compare the differences between two sets of ballistic stretching and two sets of a dynamic stretching routine on vertical jump performance. Twenty healthy male and female college students between the ages of 22 and 34 (24.8 ± 3 years) volunteered to participate in this study. All subjects completed three individual testing sessions on three nonconsecutive days. On each day, the subjects completed one of three treatments (no stretch, ballistic stretch, and dynamic stretch). Intraclass reliability was determined using the data obtained from each subject. A paired samples *t*-test revealed no significant difference in jump height, force, or power when comparing no stretch with ballistic stretch. A significant difference was found on jump power when comparing no stretch with dynamic stretch, but no significant difference was found for jump height or force. Statistics showed a very high reliability when measuring jump height, force, and power using the Kistler Quattro Jump force plate. It seems that neither dynamic stretching nor ballistic stretching will result in an increase in vertical jump height or force. However, dynamic stretching elicited gains in jump power poststretch.

KEY WORDS flexibility, vertical jump, force plate

INTRODUCTION

The vertical jump is widely accepted as a test of maximal leg power and is a common movement executed in many athletic events including basketball, volleyball, and track and field (12). It is therefore evident that an appropriate warm-up procedure aimed at increasing vertical jump performance should be evaluated. Recommendations set forth by the American College of Sports Medicine (ACSM) as well as the National Strength and Conditioning Association (NSCA) suggests a general warm-up consisting of aerobic exercises performed on either a bike or a treadmill, followed by more sport-specific movements, before any physical activity or athletic competition. A light stretching regimen consisting of static stretches held for 30 seconds for all major muscle groups is included in this recommended warm-up as well (1,8).

Stretching before performance is a common practice among athletes in hopes of increasing performance and reducing the risk of injury (17,20). Recent investigations have begun to study the acute effects that stretching has on performance, focusing primarily on static stretching as well as proprioceptive neuromuscular facilitation (PNF). Some of these studies have concluded that acute stretching neither helped nor inhibited performance (9,10,18), whereas other studies have demonstrated negative performance effects (3–5,14,16). Cumulative results indicate a negative impact of static stretching and PNF on performance; thus, there is a need for evaluating other stretching strategies for effective warm-up.

Only one study has examined the effect of ballistic stretching on vertical jump performance; it found no significant difference between static stretching or ballistic stretching on vertical jump performance in trained women (18). However, Yamaguchi and Ishii found that dynamic stretching increased leg extension power significantly in a population of healthy college males (19). Because the vertical jump is used as a test for maximal leg power, performing dynamic stretching as part of a warm-up before executing the vertical jump should lead to similar improvements as found by Yamaguchi and Ishii.

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22(6)/1844–1849

Journal of Strength and Conditioning Research
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1844 ^{the}Journal of Strength and Conditioning Research

Dynamic stretching raises core body and deep muscle temperatures, stimulates the nervous system, decreases the inhibition of antagonist muscles, increases postactivation potentiation, and possibly reduces the risk of injury (6,19). As a result of these effects, dynamic stretching may enhance force development, power development, and vertical jump performance similar to the enhancements achieved by dynamic stretching on leg extension power observed by Yamaguchi and Ishii (19). However, further research is needed to draw any conclusions of the effects of acute dynamic stretching on performance and to further explain any plausible mechanisms and/or mediating factors responsible for these effects.

It is unclear whether ballistic stretching has a positive or negative effect on performance in male subjects, and only one study has shown a negative performance effect associated with ballistic stretching in a female population (18). Because of this lack of research, further investigations on acute ballistic stretching are warranted. The purpose of this study is to compare two sets of ballistic stretching and two sets of a dynamic stretching routine on vertical jump performance. It is hypothesized that dynamic stretching will result in an increase in vertical jump height, force, and power, whereas ballistic stretching will have no significant effect on any of the three variables.

METHODS

Experimental Approach to the Problem

Stretching protocols were administered to each subject on the second and third days of testing. These protocols were designed to target major muscle groups used during a countermovement jump. Jump height, force, and power were measured using the Kistler Quattro Jump (Amherst, New York) force plate. All subjects completed three countermovement jumps on the force plate for data collection. Jump height was calculated by the force plate on the basis of the subject's height and time of flight for each jump and was expressed in centimeters (cm). Reliability of the force plate measurements were analyzed by running an intraclass reliability in the statistical analysis.

Subjects

Twenty healthy college students (10 males, 10 females), ages 21–34 years, volunteered to participate in this study. Physical characteristics of the subjects are as follow: age = 24.8 ± 3 years, height = 174.4 ± 8 cm, and weight = 81 ± 25 kg. Subjects were recruited from the general student population of a local university. After an explanation of the procedures and risks involved, all subjects signed an informed consent approved by the university's human studies committee. Once informed consent was received and understood by all subjects, they completed a medical history and physical activity readiness questionnaire.

Procedures

A counterbalanced within-subject experimental design was used for this study. The within-subject design controls for

subject variability such as individual differences in flexibility, prior stretching knowledge, jumping ability, and experience as well as gender (7). Each subject completed three testing trials on three nonconsecutive days no longer than 3 days apart (Figure 1). Subjects were instructed not to engage in lower-body exercises greater than a 17 on the RPE scale 24 hours before their test, to eliminate any potential muscle soreness or fatigue. During the initial testing session, height and weight were recorded, and leg length was measured from the greater trochanter to the lateral malleolus using a Gulick tape measure. These measurements were obtained and entered into the computer so that the force plate software could calculate jump height.

On the first day of testing, before the vertical jump test, each subject walked for 5 minutes on a treadmill at a brisk but comfortable self-selected pace as part of a general phase of the standard warm-up. This procedure was conducted on all testing days, including those that involved stretching. After walking on a treadmill, each subject completed three trials of a single countermovement jump on a Kistler Quattro Jump (Amherst, New York) force plate. The force plate measured jump height, power, and force of jump. The averages of the second and third trial scores for height, force, and power were recorded by the technician administering the test. The first trial was not included in the average because the protocol did not call for any practice jumps; therefore, the first jump of each subject may have been less of a true score compared with the second and third jumps. Each subject was then randomized to one of two conditions, dynamic or ballistic stretching.

During the second and third days of testing, subjects returned to the lab at the same time of day as their first trial and performed a series of five stretches under the supervision of a technician. The technician demonstrated each stretch and then timed the subjects as they were engaged in the stretches. After completing two sets of five ballistic stretches or dynamic stretches, each subject completed three more trials of a countermovement jump on the force plate to measure jump height, force, and power. The averages of the

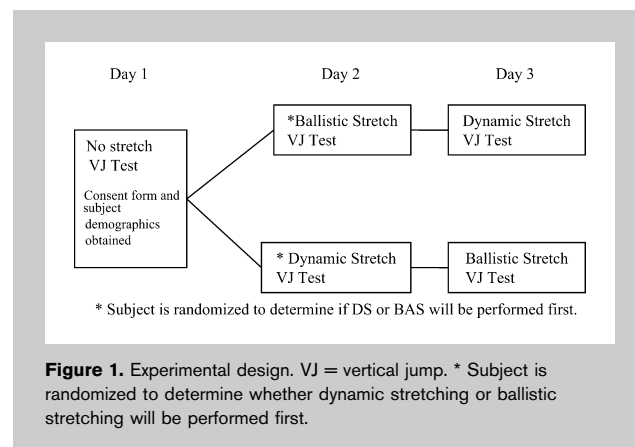


Figure 1. Experimental design. VJ = vertical jump. * Subject is randomized to determine whether dynamic stretching or ballistic stretching will be performed first.

second and third trial scores for height, force, and power were recorded. A minimum of 2 days after the second test, each subject returned to perform a third and final vertical jump test.

Stretching Protocol

The two types of stretches that were used for this study were ballistic and dynamic stretches. The ballistic stretching protocol consisted of the following five ballistic stretches targeting muscles used during a countermovement jump: forward lunge, supine knee flex, sitting toe touch, quadriceps stretch, and the butterfly. Each subject was instructed to perform the ballistic stretches by bouncing rapidly for 30 seconds at a pace of 126 beats per minute maintained by a metronome. Each subject performed two sets of the stretches. A technician timed each subject for a full 30-second stretch.

Forward Lunge. Muscles affected: Hip flexors (iliopsoas, rectus femoris).

1. Take a long step forward with the right leg and flex the right knee until it is directly over the right foot, keeping the right foot flat on the floor and the back leg straight.
2. Keep the back foot pointing toward the front foot. It is not necessary to have the back heel on the floor.
3. Keep the torso upright and rest the hands on hips or in front of the leg.
4. Bounce the hips in a forward and downward motion.
5. Repeat for the left leg.

Supine Knee Flex. Muscles affected: Hip extensors (gluteus maximus and hamstrings).

1. Lie on back with legs straight.
2. While bringing the right thigh toward the chest, flex the right knee and hip.
3. Place both hands behind the thigh and pull the thigh into the chest at a rapid, bouncing pace.
4. Repeat for the left leg.

Sitting Toe Touch. Muscles affected: Hamstrings, spinal erectors, and gastrocnemius.

1. Sit with the upper body nearly vertical and the legs straight.
2. Lean forward and attempt to grasp the toes or ankles depending on the limits of flexibility. Begin bouncing the hips, keeping the arms extended at all times.

Standing Quadriceps Stretch. Muscles affected: Quadriceps.

1. Stand upright with one hand extended out against a surface for balance.
2. Flex the right knee and raise your heel to your buttocks.
3. Grasp your right foot with one hand.
4. Pull your heel towards your buttocks and gently bounce the heel against the buttocks.
5. Repeat with the left leg.

Butterfly. Muscles affected: Hip adductors and sartorius.

1. Sitting with the upper body nearly vertical, flex both knees, bringing the soles of the feet together.

2. Pull your feet toward your body.
3. Place your hands on your feet and your elbows on your legs.
4. Bounce the knees up and down by moving the hips.

The dynamic stretching protocol consisted of five stretches that targeted the same muscle area that was stretched in the ballistic routine. Those stretches were leg kick backs, standing knee raise, calf raise, hurdle step overs, and butt kicks. These stretches were performed five times, slowly at first, and then 10 times as quickly and powerfully as possible without bouncing, for a total of 15 repetitions. Each subject completed two sets of each stretch.

Leg Kick Backs. Muscles affected: Hip flexors.

1. Stand with one foot balanced on the floor.
2. Raise the other leg slightly off the ground by bending the knee.
3. Kick the leg behind the body and fully extend the leg.
4. Repeat on the other foot.

Standing Knee Raise. Muscles affected: Hip extensors.

1. Stand with both feet firmly on the floor.
2. Raise the right leg, bringing the knee as high as possible and raising the thigh toward the chest.
3. Repeat for the left leg.

Calf Raise. Muscles affected: Soleus and gastrocnemius.

1. Start with your body in a push-up position with your feet by your side.
2. Stretch the calves by pushing one heel toward the ground, then onto the ball of the foot, and then back again.
3. Alternate between legs.

Hurdle Step Over. Muscles affected: Gluteals and hip adductors.

1. Stand with one leg slightly farther back than the other.
2. Lift the back knee high in front of the body and then rotate the leg outward and step down.
3. Repeat using the other leg.

Butt Kicks. Muscles affected: Quadriceps.

1. Stand with both feet firmly on the floor.
2. Contract the hamstrings, flex the knee, and kick the buttocks with your heel.
3. Repeat using the other leg.

Statistical Analyses

A paired samples *t*-test was used to compare mean differences between no stretch and ballistic stretch, as well as no stretch with dynamic stretch across jumping height, force, and power categories. A dependent *t*-test and repeated-measures analysis of variance were used to test whether differences in jump height, force, and/or power were affected by gender. The statistical significance was set at an alpha level of $p \leq 0.05$.

RESULTS

The means and standard deviations for each treatment group and vertical jump measurements are presented in Table 1. No subject reported any recent musculoskeletal injuries, taking any over-the-counter or prescription medications, or engaging in any lower-body/high-intensity (> 17 on rating of perceived exertion scale) training before each testing session. Gender was not found to have a significant effect on jump height, force, or power between all treatment conditions ($p = 0.339$); therefore, all results were combined before final analysis. Results indicate that there was no significant difference ($p > 0.05$) when comparing no stretch with ballistic stretch for jump height, force, or power. Similarly, there were no significant differences ($p > 0.05$) when comparing no stretch with dynamic stretch for jump height or force. However, this investigation did find a significant increase in jump power when comparing no stretch with dynamic stretch ($p = 0.023$).

Table 2 presents the intraclass reliability of the subjects' jumping performances on the force plate across three different trials. The reliability coefficients were very high: > 0.99 for jumping height, > 0.94 for jumping force, and > 0.99 for jumping power. The standard errors of measurement (*SEM*) are also presented in Table 3. The jumping height was measured to the nearest centimeter, and all standard errors were lower than 1.7 cm. These findings indicate that the Kistler Quattro Jump (Amherst, New York) force plate is a reliable tool for measuring changes in jump height, force, and power when measured at two different time points.

DISCUSSION

This study determined the effects of two stretching techniques (ballistic stretch and dynamic stretch) on maximum vertical jump height, force, and power. The results indicate that there was no significant difference in vertical jump height, force, or power when comparing no stretch with ballistic stretch. There also was no significant difference in vertical jump height or force when comparing no stretch with dynamic stretch. However, a significant difference was found in vertical jump power when comparing no stretch with dynamic stretch. Similar findings were reported by Unick

TABLE 2. Intraclass reliability ($R_{xx'}$) and standard errors of measurement (*SEM*) for the jumping height, force, and power.

Treatment	Height		Force		Power	
	$R_{xx'}$	<i>SEM</i>	$R_{xx'}$	<i>SEM</i>	$R_{xx'}$	<i>SEM</i>
No stretch	0.98	1.7	0.94	0.08	0.99	0.7
Ballistic stretch	0.98	1.4	0.97	0.04	0.99	0.55
Dynamic stretch	0.99	1.1	0.99	0.03	0.99	0.57

et al. when examining the effects of ballistic stretching as part of a warm-up routine before physical activity (18). These findings also agree with those of Yamaguchi and Ishii, who found a significant increase in the power production of the leg extensors on completion of a dynamic stretching routine for warm-up (19).

Few studies have examined the effects of ballistic stretching as part of a warm-up routine. Of those examined, only one evaluated the effects of a ballistic stretching routine on maximum vertical jump height. The finding of the current investigation is consistent with results from Unick et al. (18). They found no significant difference in maximum vertical jump height after completing three sets of a ballistic stretching routine as part of a warm-up. Both investigations used lower-body ballistic stretches that focused primarily on the quadriceps and hamstrings. However, Unick et al. used a different subject population and also incorporated a 4-minute walking period in between the stretching routine and vertical jump test. Their subjects consisted of 16 trained women college basketball players, whereas the current investigation recruited 10 males and 10 females from a more general population. In the current investigation, only three of the male subjects were currently involved in both weight training and cardiovascular training 2–4 days per week. Five of the female subjects reported involvement in weight and cardiovascular training 2–4 days per week, and one female subject currently runs track. The remaining subjects did not report current physical activity.

The reason for this neutral effect on vertical jump performance is unknown. Other than this particular investigation, Unick et al. are the only researchers to investigate the effects of ballistic stretching on performance variables. Unick et al. associated the neutral effect with the recovery of the motor neuron excitability

TABLE 1. Mean and *SD* for no stretch, ballistic stretch, and dynamic stretch with jumping height, force, and power.

	No stretch	Ballistic	Dynamic
Jump height (cm)	48 ± 12.1	50 ± 10.5	50.1 ± 12.6
Force (% body weight)	79 ± 0.26	8 ± 0.25	82 ± 0.31
Power (W·kg ⁻¹)	22 ± 7.1	22.7 ± 6.3	22.9 ± 6.8*

*Indicates significant results ($p < 0.05$).

caused by the Hoffmann reflex (H-reflex) (18). The H-reflex is similar to the spinal stretch reflex, but it is able to bypass the muscle spindle directly, making it a useful tool for measuring monosynaptic reflex activity in the spinal cord (15). The H-reflex is an estimate of alpha motor neuron excitability. This measurement can be used to assess the response of the nervous system to various neurological conditions, musculoskeletal injuries, application of therapeutic modalities, pain, exercise training, and performance of motor tasks (13). After completing 10 consecutive stretches, Avela et al. found a reduction in the H-reflex (2). This reduction in the H-reflex as well as motor neuron excitability could lead to a reduction in stretch reflex sensitivity (2). However, after reviewing the investigation of Avela et al., it is unclear whether the reduction in the H-reflex was a result of static stretches or ballistic stretches. Further research needs to be conducted to discover any neural changes within the muscles after acute ballistic stretching.

The results of this study add to the conflicting research of other studies in measuring performance variables after a stretching routine as part of a warm-up. Yamaguchi and Ishii found a significant increase in power production after completing a dynamic stretching routine in 11 healthy male subjects (19). These findings agree with those of the current investigation. A major difference between the two investigations is that Yamaguchi et al. had subjects perform only one set of dynamic stretches before testing, whereas the stretching protocol in this investigation consisted of two sets of dynamic stretches. This difference among the protocols demonstrates that one set of dynamic stretches can be just as effective as two sets.

A more recent investigation by Little and Williams found no significant difference in vertical jump height after performing a set of dynamic stretches (11). This neutral effect clearly agrees with the findings of this study. However, Little and Williams also found significant increases in speed and agility. Why would some performance variables be enhanced by dynamic stretching but others not? One explanation given by Little and Williams was that, in their study, the subjects performed the vertical jump test first, immediately after the dynamic stretch and 4-minute jogging warm-up. They conclude that the acute detrimental effects of dynamic stretching may degrade over time because they found significant increases in all other performance variables (speed, agility, and acceleration) (11). The current investigation also performed the vertical jump test immediately after the stretching protocol, which would make the findings of this investigation plausible for that explanation.

Evidence provided by Yamaguchi and Ishii (19), Little and Williams (11), and this investigation suggests that muscular power is enhanced by dynamic stretching. This finding suggests that dynamic stretching could prove to be the ideal warm-up procedure performed before physical activity and/or athletic competition. More investigations are warranted to support this conclusion.

Twenty healthy college students participated in two different types of stretching treatments to determine the stretching effects on vertical jump performance. Performance variables (height, force, and power) measured pre- and poststretch indicate no significant differences in height or force. However, there was a significant difference in power after completing two sets of dynamic stretches, but there was no significant difference in power after completing two sets of ballistic stretches.

Although the subjects for this investigation were recruited from a general population and were not necessarily athletes undergoing training, the investigators felt that this did not affect the findings. As stated in the methods section, a counterbalanced, within-subject design was used to control for subject variability. If anything, this type of design with this particular population would have resulted in a misleading significant finding with jump height because of a possible retesting and training/learning effect, which was not the case. To our knowledge, no research exists that demonstrates a difference in stretching effect on the variables measured between a trained and untrained population.

Future investigations should focus on determining any neural changes within the muscle after acute ballistic and dynamic stretching. Because this investigation used a more general subject population, it is unclear whether there would be similar findings in an athletic population. Therefore, future investigations should also look into the effects of dynamic stretching on performance variables in athletes. The acute effects of stretching on the upper body should also be looked into more extensively.

PRACTICAL APPLICATIONS

When designing a warm-up routine before physical activity, coaches and athletic trainers should limit any stretching routines to only dynamic stretches. Current evidence suggests that performing only one set of dynamic stretching is just as effective as two sets in producing a significant increase in leg power. Athletes and physically active individuals should rely mostly on quick movements that take the joint through its full range of motion when warming up. Static and proprioceptive neuromuscular facilitation should be avoided entirely when warming up before physical activity because these types of stretches have been shown to have detrimental effects.

ACKNOWLEDGMENTS

Research was conducted at the University of Louisville, Exercise Physiology laboratory. The results of this study do not constitute endorsement of the product by the authors or the NSCA.

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